

BOKARO STEEL PLANT — ILLUSTRATIVE SAMPLE

Standard Operating Procedure

SOP No: BSP-SAF-009

Title: Confined Space Entry

Department: Safety and Environment

Revision: 04

Date: 03-Jan-2024

1. PURPOSE

To prevent injury or death from toxic gases, oxygen deficiency, engulfment or entrapment when entering confined spaces.

2. SCOPE

Applies to all confined spaces including gas mains, dust catchers, bunkers, tanks, pits, sewers, stoves and ducts. Applies to employees and contractors.

3. PERMIT REQUIREMENT

Entry is allowed only with a valid Confined Space Entry Permit issued under the Permit-to-Work system (BSP-SAF-014). The permit is valid for one shift only.

4. ATMOSPHERIC TESTING (BEFORE ENTRY AND EVERY 2 HOURS)

- Oxygen: between 19.5% and 23.5%
- Carbon monoxide (CO): below 25 ppm
- Hydrogen sulphide (H₂S): below 10 ppm
- Flammable gas: below 10% of the Lower Explosive Limit (LEL)

Test at the top, middle and bottom of the space with a calibrated multi-gas detector.

5. ISOLATION

- Positive isolation of all gas, steam, water and material lines by blinding or removing a spool piece
- Electrical and mechanical isolation with LOTO as per BSP-EL-005
- Purge with nitrogen and then air until readings are within limits

6. ENTRY RULES

- A trained attendant (hole watcher) must remain at the entry point at all times and must never enter
- Maximum 2 persons inside unless the permit states otherwise
- Continuous forced ventilation during the work
- Full body harness with lifeline for vertical entry
- 24 V or lower hand lamps only

7. EMERGENCY RESCUE

- Rescue only by persons wearing SCBA
- A rescue tripod and winch must be kept ready at the entry point for vertical spaces
- Call Emergency Control Room: Ext. 100